Unit 2, Lesson 5: Rational and Irrational Numbers on the Number Line

Benchmark

MA.8.NSO.1.2 Plot, order, and compare rational and irrational numbers, represented in various forms.

Learning Targets

- ☐ I can compare rational and irrational values.
- ☐ I can order rational and irrational values.
- ☐ I can plot rational and irrational numbers on a number line.

2.5.1 Warm-Up: Which One Doesn't Belong?

1. Four numbers are shown in the table. Choose **two** of the values to write an argument for why each does not belong with the others.

Number	Why doesn't this belong with the others?
3.14	
$\frac{22}{7}$	
π	
$3.\overline{14}$	

2.5.2 Guided Instruction: Compare and Order Rational and Irrational Numbers

1. Use the numbers shown to complete the following.

$0.125 \qquad -\sqrt{12} \qquad -8 \qquad -\sqrt{49} \qquad -2\pi \qquad \sqrt[3]{-8} \qquad -\frac{13}{3}$

a. Write each number under the appropriate description in the table.

Rational Numbers	Irrational Numbers

b. Order the numbers from least to greatest.

c. Plot and label each number on the number line.

-10 -9 -8 -7 -6 -5 -4 -3 -2 -1 0 1 2

2. Complete the table by comparing each pair of expressions using >, <, or =. Then, plot and label the expressions on the number line.

Compare (>, <, or =)	Number Line Plot
$\sqrt{36}$ ○ < ○ > ○ = $\sqrt[3]{36}$	-1 0 1 2 3 4 5 6 7 8
$\sqrt{89}$ ○ < ○ > ○ = 9.89	4 5 6 7 8 9 10
$\frac{9}{2}$ ○ < ○ > ○ = $\sqrt{23}$	4 5 6 7 8 9 10
$\pi + 3$ ○ < ○ > ○ = 2π	-1 0 1 2 3 4 5 6 7 8

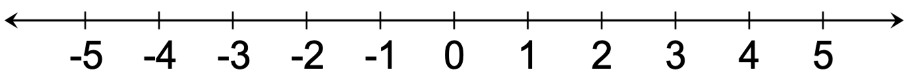
2.5.3 Guided Practice: Compare and Order Rational and Irrational Numbers

1. Use the numbers shown to complete the following.

$$-4.5 \qquad \frac{3}{4} \qquad \sqrt{7} \qquad 3\,\frac{2}{3} \qquad -\sqrt[3]{64} \qquad \frac{\pi}{3}$$

   a. Order the numbers from greatest to least.

   b. Plot and label each number on the number line.



2.5.4 Activity: Ordering Rational and Irrational Numbers

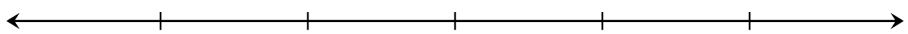
1. Your teacher will give you a card with a numerical expression on it.

   a. What is the numerical expression on your card?

   b. Does the expression include any irrational numbers?

   c. If your expression contains only rational numbers, write the exact value of your card, expressed as a decimal. If your expression contains any irrational numbers, write the approximate value of your card, expressed as a decimal.

2. Once in your group, label the tick marks on the number line to match your group's assigned number line.



   a. Work with your group to order your number cards from least to greatest.

   b. Work with your group to determine the approximate location of each expression on the number line. Then, plot and label each expression on your number line.

2.5.5 Bring It Together: Ordering Rational and Irrational Numbers

1. Use your experience doing the previous activity to complete the following.

   a. List any strategies you learned that are helpful to keep in mind when comparing and ordering rational and irrational numbers or plotting them on a number line.

Strategies for Comparing and Ordering	Strategies for Plotting on a Number Line

   b. Compare and discuss your list with your partner. Add any strategies your partner shares that you find helpful that aren't on your list.

2.5.6 Wrap-Up: Cool Down

1. Write a summary of what you learned in this lesson.